

Individual assignment 5

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Set up

Packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# wrap axis labels
library(scales)

# reading .xlsx files
library(readxl)

# arrange plots
library(patchwork)
```

Data

```
# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
```

```

    sheet = "Aquatic Insects"
  )

# site information
sites <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites"
)

# water quality
water_quality <- read_xlsx(
  here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Water Quality",
  na = c("", "n/a", "N/A", "over", "173+")
)

```

Cleaning

```

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")

```

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

```

```

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO

```

```

summarize(med_ph = median(p_h, na.rm = TRUE),
          med_sal = median(salinity_ppt, na.rm = TRUE),
          med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE),
          .groups = "drop") |>
# ungroup data frame
ungroup() |>
# filter out salinity outliers
filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
# clean column names
clean_names() |>
# filtering join: only include sites occurring in ncos_sites
semi_join(ncos_sites, by = c("site")) |>
# renaming columns
rename(date = date_on_vial) |>
# select columns of interest
select(date, sample_type, site,
       ostracod,
       copepod,
       hemiptera_corixidae_boatman,
       cladocera,
       nematode,
       diptera_ceratopogonidae,
       annelida_oligochaete,
       diptera_chironomid,
       annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
             names_to = "taxon",
             values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter
summarize(
  ave_lit = sum(abundance, na.rm = TRUE) / 7.5,
  .groups = "drop"
) |>
# ungroup data frame

```

```

ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(
  site_full, med_sal, .fun = "median", .na_rm = TRUE
),
  site_full = fct_rev(site_full))

```

Class visualizations

```

# base layer
salinity <- ggplot(data = aq_ins_clean,
  # x-axis: site
  # y-axis: mean salinity
  mapping = aes(x = site_full,
    y = med_sal)) +
# first layer: salinity measurements at each survey
geom_jitter(height = 0,
  width = 0.1,
  shape = 21) +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(width = 15)) +
# second layer: points to represent means
stat_summary(fun = median,

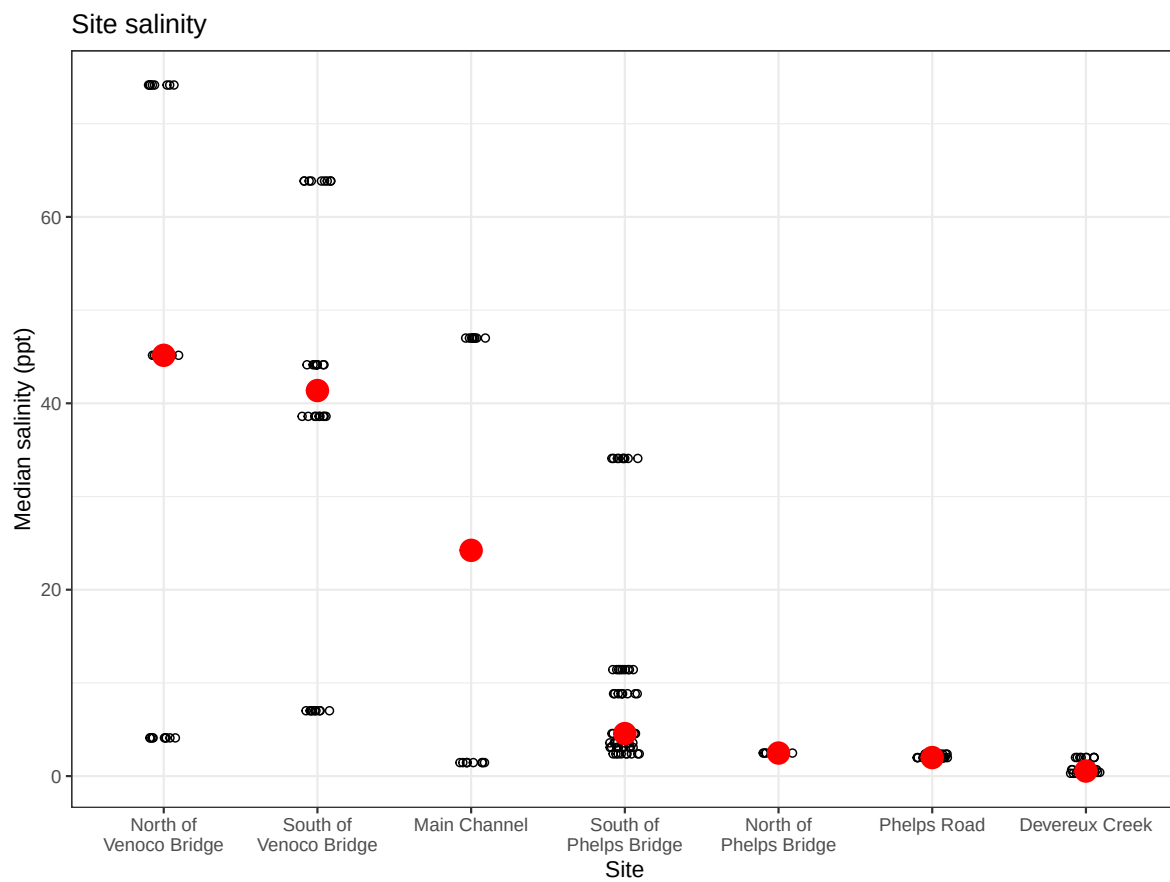
```

```

        color = "red",
        size = 1) +
# relabelling x- and y-axes
labs(x = "Site",
     y = "Median salinity (ppt)",
     title = "Site salinity") +
# different theme
theme_bw()

# displaying object
salinity

```



```

# creating new object from clean aquatic inverts
copepods <- aq_ins_clean |>
# filter to only include copepods
filter(taxon == "copepod") |>

```

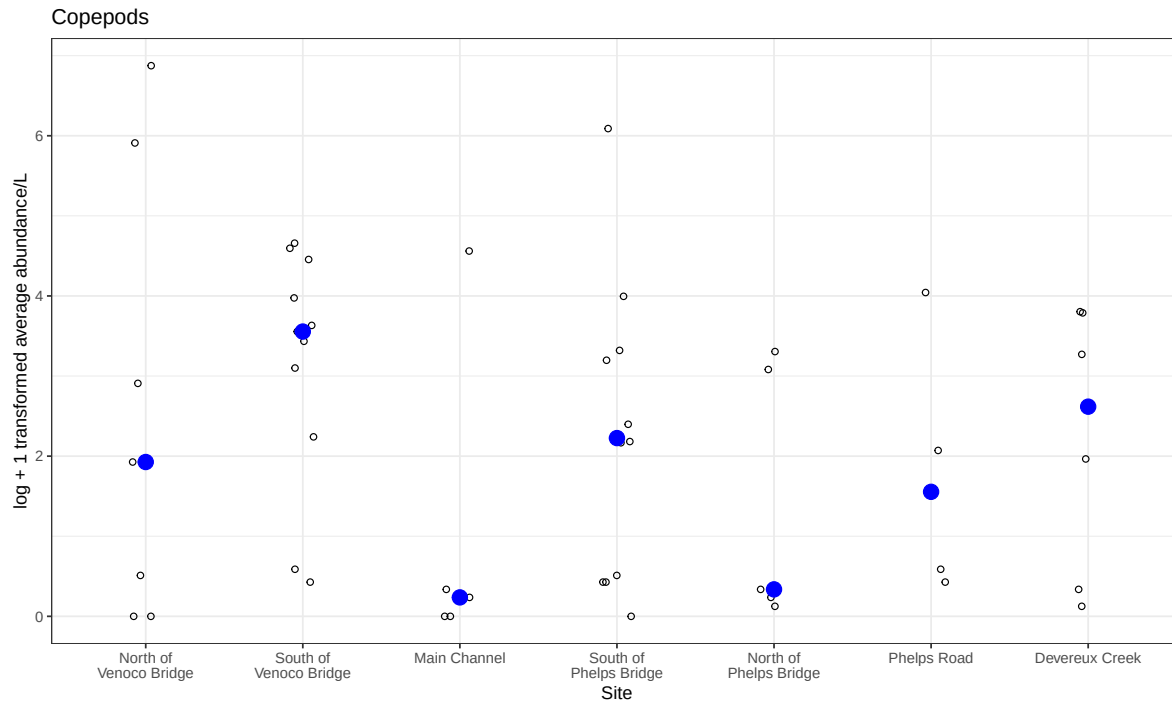
```

# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1))

# base layer
copepods <- ggplot(data = copepods,
  # x-axis: site
  # y-axis: mean abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
# first layer: points to represent observations
geom_jitter(height = 0,
  shape = 21,
  width = 0.1) +
# second layer: points to represent medians
stat_summary(geom = "point",
  fun = "median",
  size = 4,
  color = "blue") +
# wrapping x-axis text
scale_x_discrete(labels = label_wrap(15)) +
# relabelling x- and y-axis, adding title
labs(x = "Site",
  y = "log + 1 transformed average abundance/L",
  title = "Copepods") +
# different theme
theme_bw()

# displaying object
copepods

```



1. Visualizing differences across sites in major taxon abundance

a.

- x-axis: site
- y-axis: log + 1 transformed average ostracod abundance/L
- geometry to represent average abundance/L: `geom_jitter()`
- geometry to represent medians: `stat_summary()` with a visually distinct point

b.

```
# creating new object from clean aquatic inverts
ostracods <- aq_ins_clean |>
  # filter to only include ostracods
  filter(taxon == "ostracod") |>
  # create log + 1 transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1))
```

```
# display 10 random rows from data frame
slice_sample(ostracods, n = 10)
```

```
# A tibble: 10 x 24
```

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2023-05-04_NMC	2023-05-04 00:00:00	NMC	ostr~	1.87e+0	8.32	NA	10.3
2	2023-02-28_NPB	2023-02-28 00:00:00	NPB	ostr~	2.53e+0	NA	NA	NA
3	2022-08-12_NVBR	2022-08-12 00:00:00	NVBR	ostr~	0	NA	NA	NA
4	2020-01-21_NDC	2020-01-21 00:00:00	NDC	ostr~	2.4 e+0	6.53	0.3	7.6
5	2023-05-16_NVBR	2023-05-16 00:00:00	NVBR	ostr~	1.33e-1	9.29	NA	9.83
6	2023-05-05_NPB	2023-05-05 00:00:00	NPB	ostr~	2.47e+1	NA	NA	12.1
7	2023-08-12_NEC	2023-08-12 00:00:00	NEC	ostr~	0	NA	NA	NA
8	2023-02-12_NEC	2023-02-12 00:00:00	NEC	ostr~	2.13e+0	NA	NA	NA
9	2023-11-11_NEC	2023-11-11 00:00:00	NEC	ostr~	0	NA	38.6	0.25
10	2022-05-12_NPB2	2022-05-12 00:00:00	NPB2	ostr~	1.20e+3	NA	NA	NA

```
# i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
#   Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
#   Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
#   comments <chr>, site_full <fct>, log_ave_lit <dbl>
```

c.

```
# base layer
ostracods_plot <- ggplot(data = ostracods,
  # x-axis: site
  # y-axis: log transformed average abundance/L
  mapping = aes(x = site_full,
    y = log_ave_lit)) +
  # first layer: points to represent individual observations
  geom_jitter(height = 0,
    width = 0.1,
    shape = 21) +
  # second layer: larger colored diamonds to represent site medians
  stat_summary(geom = "point",
    fun = "median",
    size = 4,
    shape = 23,
    color = "darkorange") +
  # wrap long site names so x-axis labels are readable
```

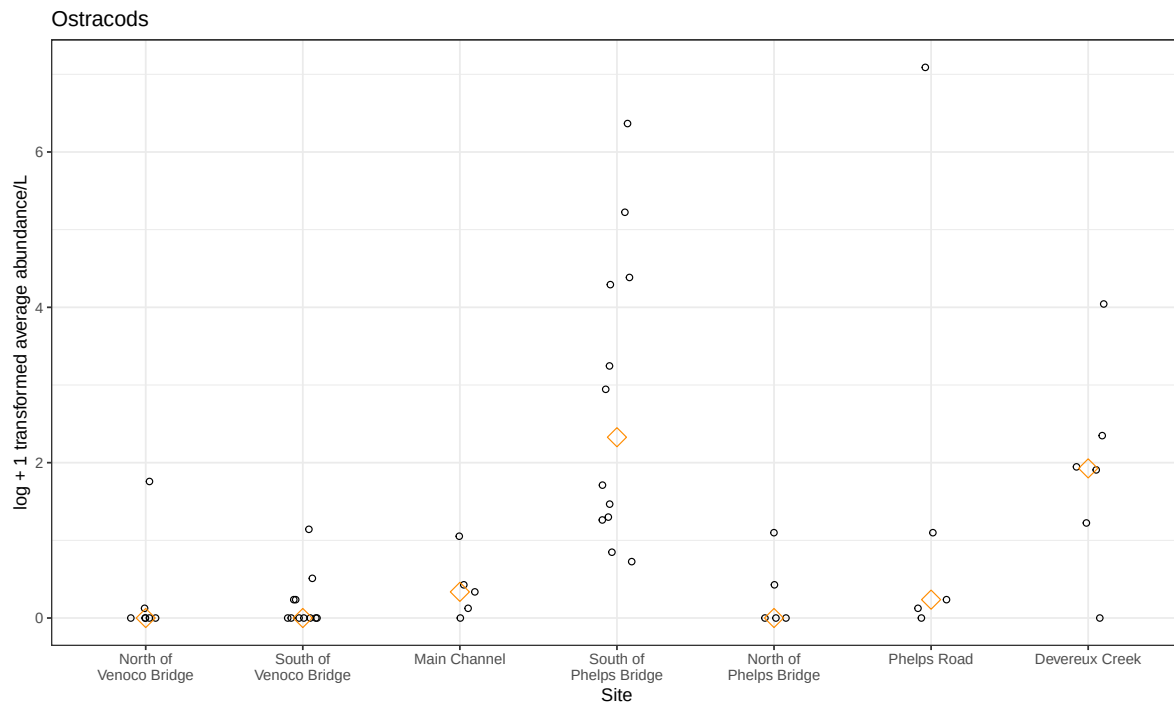


```

scale_x_discrete(labels = label_wrap(width = 15)) +
# relabel axes and add a descriptive title
labs(x = "Site",
      y = "log + 1 transformed average abundance/L",
      title = "Ostracods") +
# use a non-default theme
theme_bw()

# display plot
ostracods_plot

```

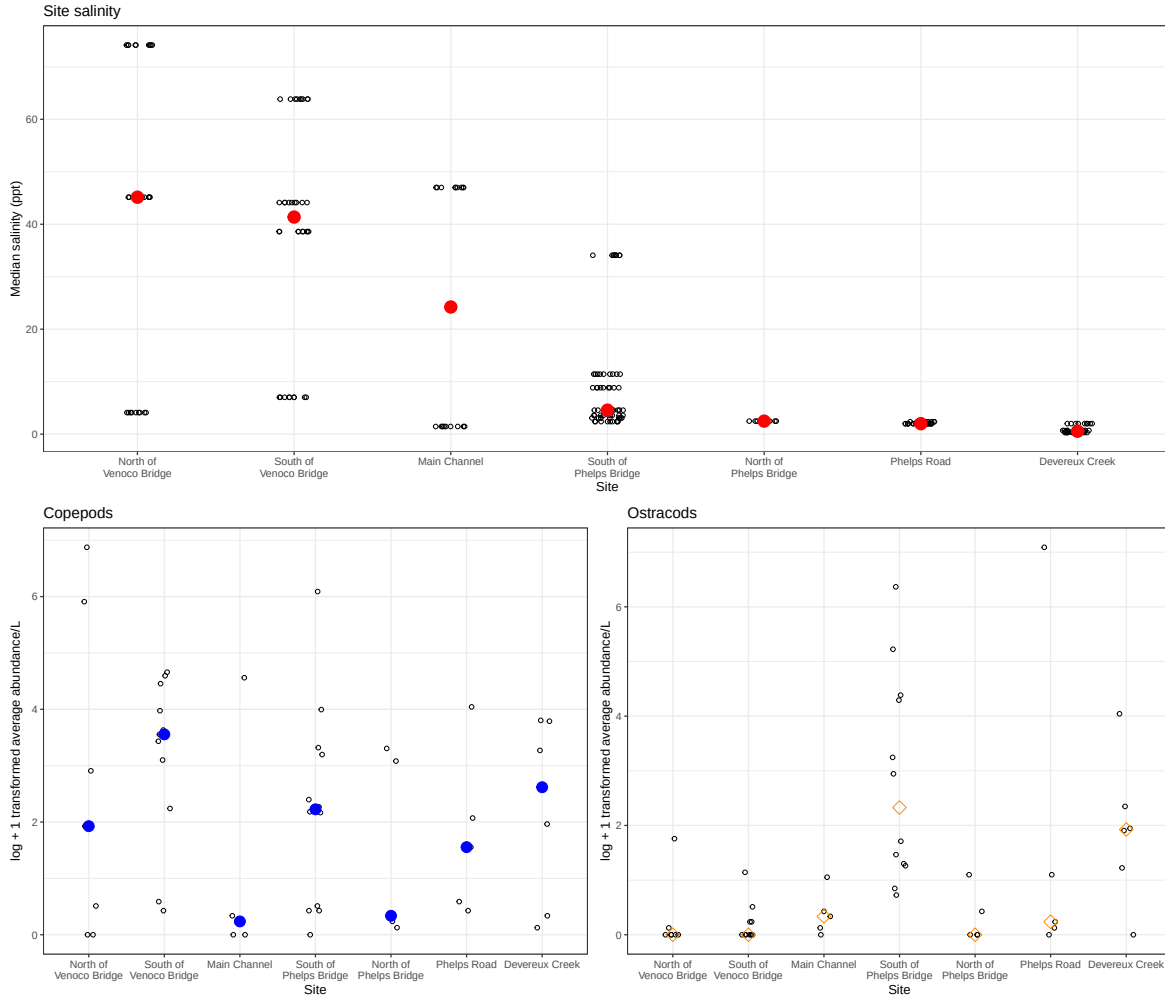


d.

```

# combine salinity, copepod, and ostracod plots
salinity / (copepods | ostracods_plot)

```



e.

Median copepod abundance is highest at South of Venoco Bridge and Devereux Creek, while median ostracod abundance is highest at South of Phelps Bridge and Devereux Creek. This is shown by the larger colored median points in each panel, as well as the spread of individual observations around those medians.

Ostracod abundance does not appear to closely follow the salinity gradient across sites. For example, South of Phelps Bridge has relatively low median salinity in the top panel but the highest median ostracod abundance in the bottom-right panel, suggesting ostracods may be less strongly associated with salinity than copepods.

2. Visualizing total abundance as a function of salinity

a.

- x-axis: median salinity, `med_sal`
- y-axis: log + 1 transformed average abundance/L, `log_ave_lit`
- color: taxon
- geometry: `geom_point()`
- facets: `facet_wrap(~ taxon, scales = "free")`

b.

```
# create data frame for abundance-salinity figure
abundance_salinity <- aq_ins_clean |>
  # create log + 1 transformed abundance column
  mutate(log_ave_lit = log(ave_lit + 1)) |>
  # change taxon names from snake case to title case
  mutate(taxon = str_to_title(str_replace_all(taxon, "_", " "))) |>
  # order taxon levels by maximum average abundance/L
  mutate(taxon = fct_reorder(taxon, ave_lit, .fun = max, .desc = TRUE)) |>
  # keep columns needed for plot
  select(taxon, med_sal, ave_lit, log_ave_lit)

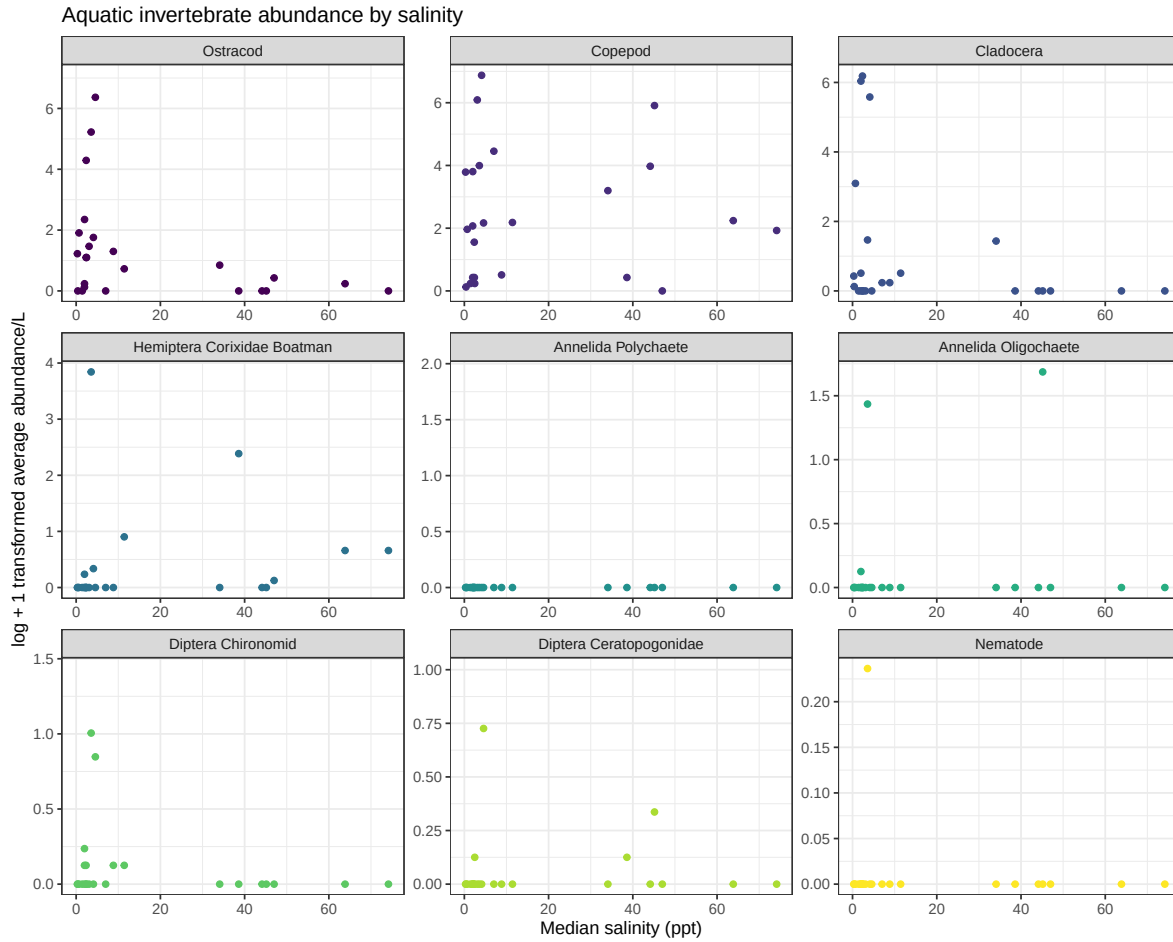
# display 10 random rows from data frame
slice_sample(abundance_salinity, n = 10)
```

A tibble: 10 x 4

	taxon	med_sal	ave_lit	log_ave_lit
	<fct>	<dbl>	<dbl>	<dbl>
1	Hemiptera Corixidae Boatman	45.2	0	0
2	Ostracod	NA	79.2	4.38
3	Copepod	2.47	0.267	0.236
4	Nematode	NA	0	0
5	Diptera Ceratopogonidae	1.98	0	0
6	Diptera Chironomid	NA	0.133	0.125
7	Cladocera	NA	0	0
8	Hemiptera Corixidae Boatman	NA	0	0
9	Diptera Chironomid	2.36	0.133	0.125
10	Annelida Oligochaete	11.4	0	0

C.

```
# base layer
ggplot(data = abundance_salinity,
       # x-axis: median salinity
       # y-axis: log transformed average abundance/L
       # color: taxon
       mapping = aes(x = med_sal,
                     y = log_ave_lit,
                     color = taxon)) +
# points show each taxon's abundance at each salinity value
geom_point() +
# separate each taxon into its own panel with free axes
facet_wrap(~ taxon,
          scales = "free") +
# use custom colors instead of ggplot defaults
scale_color_viridis_d() +
# relabel axes and add title
labs(x = "Median salinity (ppt)",
     y = "log + 1 transformed average abundance/L",
     title = "Aquatic invertebrate abundance by salinity") +
# use a non-default theme
theme_bw() +
# remove legend because taxa are already labeled by facet titles
theme(legend.position = "none")
```



d.

Across taxa, abundance appears to vary more strongly at low salinity than at high salinity. Ostracods, cladocerans, diptera chironomids, and annelida oligochaetes have their highest log-transformed abundance values at low salinity, with abundance generally closer to zero at higher salinity values. Copepods show less of a clear negative relationship because some high abundance values also occur at moderate to higher salinity levels. Several taxa, including nematodes, diptera ceratopogonidae, and annelida polychaetes, have mostly low abundance across the salinity range, making their relationship with salinity harder to interpret.